# Aura: A Neural-Physics-Based Routine Scheduling System for Urban UAV Delivery Networks

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**Abstract**

We present *Aura*, a UAV routine scheduling system for urban delivery networks built on a station-to-station operational model. The city is partitioned into neighborhood units, each served by a drone base station whose capacity scales with local population density. All UAV operations—delivery, data collection, and hazard response—are unified under a single abstraction: a flight segment from station $A$ to station $B$. A time-varying global energy field governs both fleet rebalancing across stations (macro scale) and individual trajectory execution (micro scale), drawing on analogies with $r$-adaptive mesh methods and multi-physics particle dynamics. The airspace is organized as a *trunk-and-branch* hierarchy: trunk corridors connect high-demand station pairs with precomputed nominal trajectories and narrow tolerance tubes, while branch corridors provide city-wide coverage with wider tubes optimized for opportunistic sensing. Nominal trajectories are computed once via anisotropic Fast Marching on a SCALED-driven 4D wind cost field and incrementally updated as the underlying environment evolves; in-flight execution remains onboard via robust tube-following MPC. This separation of *offline routing infrastructure* from *onboard execution* mirrors all webscale routing systems (BGP, CDN, OSM-routing) and is the prerequisite for city-scale deployment. We outline a three-stage simulation roadmap, beginning in 20 km$^2$ of Nanshan District (Shenzhen) and culminating in city-wide deployment.

## 1 Introduction

The rapid growth of urban aerial logistics demands a scheduling framework that can simultaneously handle heterogeneous task types—package delivery, aerial data collection, and emergency response—while respecting physical constraints such as wind, obstacles, noise regulations, and airspace policy. This document describes the *Aura* system, whose key insights are (i) all UAV missions can be reduced to station-to-station flight segments within a base-station network, enabling a unified energy-field formulation for both fleet-level allocation and vehicle-level path planning; (ii) the airspace is organized as a hierarchical *trunk-and-branch* network, with offline-precomputed nominal trajectories and online tube-following execution; and (iii) the same fleet serves transport and sensing through time-shifted dual-mode operation, closing a self-improving loop between physical execution and the underlying neural surrogate models.

**Architectural commitments.** The Aura architecture rests on six commitments that together fully specify the system from network topology down to onboard control:

- **C1. All-pairs precomputed nominal trajectories** via anisotropic Fast Marching on a SCALED-driven 4D wind cost field.
- **C2. Tube-tolerance execution** with onboard robust Model Predictive Control (MPC).
- **C3. Incremental field-driven trajectory updates** triggered by environmental change detection (geometry / wind / regulatory).
- **C4. Self-improving fleet-sensing loop** via opportunistic onboard sensing during operational missions.
- **C5. Trunk-and-branch hierarchical topology** with dual-purpose network design.
- **C6. Time-shifted dual-mode fleet operation** (transport / sensing) with information-gain-based dispatch.

Each subsequent section develops one or more of these commitments and identifies the existing algorithms that instantiate them.

## 2 Operational Model: Station-Based Urban Delivery Network

The city is divided into *neighborhood* units, each equipped with a UAV base station. The physical scale of each station—number of landing pads, buffer capacity, and platform area—is determined by the population density and community concentration of its neighborhood: denser, higher-demand areas receive larger stations; sparser areas receive smaller ones.

Stations are sited at *street centroids* of the OSM road graph rather than on building rooftops or at dedicated vertiports. This choice (i) inherits road network topology as the routing skeleton, (ii) leverages street canyons as semi-protected vertical takeoff/landing corridors, (iii) decouples station siting from building ownership negotiations, and (iv) places stations in publicly governed space (consistent with regulatory expectations). The trade-off—canyon vortex effects at low altitude and pedestrian/vehicle deconfliction during takeoff/landing windows—is handled by SCALED-resolved local flow fields and station-level slot scheduling, respectively.

Under this architecture, ground-level human labor is minimized. Each neighborhood requires only 2–3 ground couriers, whose sole responsibility is to collect items from local merchants and courier pickup points and deliver them to the neighborhood base station. The station then orchestrates UAV sorties for medium- to long-range, cross-station deliveries.

The same logic extends to logistics warehouses and residential communities:

- Humans handle the *last mile*: aggregating nearby parcels/food orders into the base station.
- Residents can use the station in reverse for outbound shipments.

The resulting operational paradigm can be understood as an upgrade of the current *Cainiao Station + point-to-point courier / delivery robot* model—humans perform short-range aggregation while UAVs execute medium- to long-range cross-neighborhood transport.

**Key abstraction.** Under this model, every UAV mission reduces to a flight from **Station** $A$ to **Station** $B$. Regardless of whether the semantic task is delivery, data collection, or emergency response, the physical operation is always a single station-to-station segment. This uniform abstraction is the prerequisite for both the energy-field formulation (Section 4) and the precomputed-trajectory architecture (Section 3).

## 3 Network Topology: Trunk-and-Branch Hierarchy

A naive design would treat every station pair $(A, B)$ symmetrically. In practice, urban demand is heavy-tailed: a small fraction of station pairs accounts for the majority of traffic. Aura exploits this structure with an explicit two-tier topology, mirroring the trunk-and-branch organization observed in mature transportation and telecommunications networks (highways and arterials, BGP backbone and ISP networks, mainline rail and feeder lines).

**Trunk corridors.** Trunk corridors connect high-demand station pairs (or station clusters via hub aggregation). They have:

- **Narrow tolerance tubes** (high airspace utilization);
- **Multiple flight levels** with directional separation (no head-on traffic at the same level);
- **Strict scheduling** with reserved time-window slots;
- **Throughput-optimized cost weighting** in trajectory computation;
- **Tight regulatory binding**—trunk corridor topology is registered with airspace authorities and changes only through formal amendment cycles.

**Branch corridors.** Branch corridors extend from trunk junction nodes to lower-demand stations and to regions of the city not directly served by trunks. They have:

- **Wider tubes** (more in-flight tolerance, lower utilization but more flexible);
- **Single flight level** with bi-directional traffic at low volume;
- **On-demand activation** rather than continuous reservation;
- **Coverage- and information-gain-weighted cost** in trajectory computation;
- **Lighter regulatory binding**—branches can be reconfigured at faster cadence (weekly to monthly).

Branch attachment to trunk follows a Voronoi rule: each non-trunk station connects to its nearest trunk junction by default; this can be relaxed to k-nearest or replaced by a learned attachment policy when richer demand data is available.

**Dual purpose.** Trunk's primary purpose is efficient transport; branch's primary purpose is city-wide *coverage*—both for serving long-tail destinations and for sensing. Critically, the same fleet serves both: high-demand transport drones on trunk are periodically reassigned to branch missions under the dispatching policy of Section 9, eliminating the need for a separate sensing fleet.

**Why hierarchy is the attractor.** We emphasize that trunk-and-branch is not a compromise between free-flight and fully-structured corridors; it is the topology that all heavy-tailed demand networks evolve into [20]. Choosing it explicitly (rather than discovering it post-hoc) lets Aura inherit the engineering and regulatory practices already developed for analogous networks.

# 4 Three Task Types and the Global Energy Field

Urban UAV tasks are classified into three categories:

1. **Delivery** — package and food delivery between stations.
2. **Data collection** — aerial sensing, mapping, and monitoring.
3. **Hazard assistance** — emergency response and disaster relief.

The global state of the city is described by a time-varying *energy field*:

$$E(\mathbf{x}, t) \;=\; E_{\text{delivery}}(\mathbf{x}, t) \;+\; E_{\text{data}}(\mathbf{x}, t) \;+\; E_{\text{hazard}}(\mathbf{x}, t) \;+\; \text{BC}, \tag{1}$$

where the boundary conditions BC encode wind fields (from SCALED), building geometry (from the urban geometry pipeline), and regulatory no-fly zones—i.e., the domain itself is jointly defined by physics and policy. The field evolves over time: new tasks inject energy; completed tasks dissipate it.

Each UAV operation is a station-to-station flight:

- **Delivery** is the most direct case.
- **Data collection** corresponds to a detour budget along the route, expressed as a soft attractor toward high-uncertainty regions of the SCALED field.
- **Hazard assistance** manifests as a high-priority spike that pulls nearby UAVs toward the incident.

This energy field is optimized simultaneously at two scales: macro-scale fleet allocation (Section 4.1) and micro-scale trajectory execution (Section 5).

## 4.1 Macro Scale: UAV Density at Each Base Station

The number of UAVs allocated to each station should be proportional to the integral of $E$ over that station's service area, subject to the station's physical capacity (determined by neighborhood population density and community concentration).

This is a literal translation of **$r$-adaptive mesh methodology** as developed in the GMORPH line of work [21]:

$$\begin{aligned} \text{Base station} &\;\longleftrightarrow\; \text{Grid node} \\ \text{UAV count} &\;\longleftrightarrow\; \text{Degrees of freedom (DoF)} \\ E &\;\longleftrightarrow\; \text{Monitor function} \end{aligned}$$

The diffusion equation that redistributes grid density under the monitor function directly corresponds to inter-station fleet rebalancing. Fleet-size conservation,

$$\sum_k n_k \;=\; N, \tag{2}$$

is the $r$-adaptivity constraint of "no new degrees of freedom." In practice, the rebalancing diffusion is solved on the network graph rather than a continuous domain, but the variational principle is identical.

At this scale, all three task types contribute to $E$ simultaneously: delivery sets the baseline density; data collection contributes a slowly varying background term; and emergency events appear as instantaneous spikes triggering reallocation.

**Rebalancing logic.** The system computes the *expected* UAV density $n_k^*$ at each station. When $n_k^* > n_k$ (expected demand exceeds current supply), the system prioritizes dispatching inbound UAVs from neighboring stations via delivery orders routed through this station. When $n_k^* < n_k$, the station reduces its fleet by dispatching outbound UAVs. This mechanism prevents the scenario where a high-demand station is starved of UAVs.

**Mission assignment (L0.5).** Once fleet density is set, individual missions must be assigned to specific drones at specific stations under time-window, battery, and pickup-delivery constraints. This is a stochastic, multi-depot, electric Pickup-and-Delivery Problem (PDP) with weather-dependent edge costs. We adopt a three-tier solver:

- **Strategic (daily/weekly):** Two-stage stochastic Mixed Integer Programming for fleet sizing and shift planning, solved via Gurobi/CPLEX.
- **Tactical (minute-scale):** Adaptive Large Neighborhood Search (ALNS) [14] on the current order book, with SCALED-aware edge costs.
- **Reactive (second-scale):** A learned policy (RL trained on ALNS solutions, in the spirit of [15]) handles in-flight re-assignment under sudden order arrival or weather changes.

# 5 Micro Scale: Nominal Trajectory and Tube-Tolerance Execution

The defining architectural decision at the micro scale is the *separation of offline routing infrastructure from onboard execution*. This separation is what makes city-scale operation tractable; it mirrors the design of every webscale routing system (BGP for the Internet, CDN edge routing, OSM-routing for ground transportation), all of which precompute routing tables offline and execute lookups onboard rather than re-solving routing per packet/request.

## 5.1 Offline Layer: All-Pairs Nominal Trajectories

For each station pair $(A, B)$, Aura precomputes a *nominal trajectory* $\pi_{AB}^*$: the time-parametrized 3D path from $A$ to $B$ that minimizes the multi-objective cost

$$J(\pi) = \int_0^T \Big[ w_e\, c_{\text{energy}}(\mathbf{x}, t) + w_n\, c_{\text{noise}}(\mathbf{x}) + w_r\, c_{\text{risk}}(\mathbf{x}) + w_c\, c_{\text{congestion}}(\mathbf{x}, t) \Big]\, ds, \tag{3}$$

subject to building, regulatory, and dynamic feasibility constraints. The cost field is provided directly by the SCALED 4D wind surrogate (for $c_{\text{energy}}$), the urban geometry pipeline (for $c_{\text{risk}}$ and feasibility), and the population/sensitivity layer (for $c_{\text{noise}}$).

**Algorithm: anisotropic Fast Marching.** Computing nominal trajectories reduces to solving the anisotropic Eikonal equation

$$\|\nabla T(\mathbf{x})\|_{M(\mathbf{x},t)} = 1, \qquad T(\mathbf{x}_A) = 0, \tag{4}$$

where $T(\mathbf{x})$ is the minimum-cost arrival time from $A$, and $M(\mathbf{x}, t)$ is the local cost metric tensor (anisotropic because wind and corridor preferences induce direction-dependent costs). We solve (4) with the **Anisotropic Fast Marching Method** [2, 3], which has $O(N \log N)$ complexity in the number of grid nodes and yields the trajectory $\pi_{AB}^*$ as a single-source-to-all-targets solution—all $|B|$ destinations from a given source $A$ are obtained from one Eikonal solve. This methodological choice directly extends the well-trajectory optimization framework of Lyu et al. [1], which applied Fast Marching to anisotropic subsurface reservoirs; Aura adapts the same machinery to the urban airspace cost field.

For $|S|$ stations, the all-pairs nominal trajectory database requires $O(|S|)$ FMM solves and $O(|S|^2)$ trajectories. In the small-$|S|$ regime ($|S| \lesssim 10^3$), we materialize all pairs explicitly. In the large-$|S|$ regime, we revert to a hierarchical backbone: stations are routed via their nearest trunk junction, reducing storage to $O(|S|)$ trunk-attachment trajectories plus $O(|S_{\text{trunk}}|^2)$ trunk-trunk trajectories.

## 5.2 Tube-Tolerance Layer: In-Flight Execution

Around each nominal trajectory $\pi^*_{AB}$, Aura defines a *tolerance tube*

$$\mathcal{T}_{AB}(t) = \{\mathbf{x} \in \mathbb{R}^3 : \|\mathbf{x} - \pi^*_{AB}(t)\| \leq \epsilon_{AB}(\mathbf{x}, t)\}, \tag{5}$$

where $\epsilon_{AB}(\mathbf{x}, t)$ is a time- and space-varying tube radius, narrow on trunks and wider on branches, and growing with SCALED's local wind uncertainty (so tubes are wider where the environment is harder to predict). UAVs are required to remain within $\mathcal{T}_{AB}$ during flight but are otherwise free to choose their instantaneous position—this in-tube freedom is what carries the local potential-field dynamics described in Section 5.3.

**Algorithm: tube-following robust MPC.** Each UAV runs a robust **Tube Model Predictive Controller** [4, 5] onboard. The controller solves a finite-horizon optimization problem each control cycle (typically 10–50 ms) that tracks $\pi^*_{AB}$ subject to (i) staying within $\mathcal{T}_{AB}$ under bounded wind disturbance, (ii) collision avoidance with nearby UAVs, and (iii) the local potential gradient (Section 5.3). Formal guarantees follow from **funnel synthesis** [7, 6]: under disturbances bounded by the SCALED uncertainty envelope, the closed-loop trajectory is provably contained within the tube.

## 5.3 In-Tube Particle Dynamics

Within the tolerance tube, a UAV is modeled as a *particle* in a multi-physics potential landscape:

$$m\,\ddot{\mathbf{x}} \;=\; \mathbf{F}_{\text{nominal}} \;-\; \nabla U_{\text{data}} \;-\; \nabla U_{\text{wind}} \;-\; \nabla U_{\text{geom}} \;+\; \sum_j \mathbf{F}^{\text{rep}}_{ij}, \tag{6}$$

where:

- $\mathbf{F}_{\text{nominal}}$ — strong attractor toward the current point on the nominal trajectory $\pi^*_{AB}(t)$. *This replaces the original "attractor toward destination" formulation:* the destination is reached by following the precomputed path, and only deviations from it are penalized locally.
- $-\nabla U_{\text{data}}$ — data-demand pull: within the tube, the UAV is gently attracted toward high-uncertainty regions of the SCALED field, enabling opportunistic sensing without leaving the tube.
- $-\nabla U_{\text{wind}}$ — local wind-field gradient, served by SCALED at runtime.
- $-\nabla U_{\text{geom}}$ — building repulsion, served by the geometry pipeline.
- $\mathbf{F}^{\text{rep}}_{ij}$ — soft inter-UAV repulsion, with interaction range set by the effective radius (acoustic, safety, physical envelope; see below).

The effective per-UAV radius is jointly determined by:

- **Acoustic footprint** — minimum acceptable distance from noise-sensitive areas (residential zones, hospitals, schools).
- **Minimum safety separation** — inter-UAV clearance, scaled by SCALED-predicted local turbulence.
- **Physical envelope** — the vehicle's own dimensions.

**Why this formulation works.** The trajectory *within* the tube emerges from the potential superposition—this is exactly the picture originally proposed for global trajectory planning—but the *macro path* is locked to the precomputed nominal. The net effect is that the original emergent-particle model now operates as the in-tube refinement layer of a structurally-bounded routing system, gaining all the auditability, regulatory tractability, and city-scale tractability of precomputed routing while losing none of the local responsiveness to wind, obstacles, and neighbors that motivated the particle model in the first place.

Nearest-neighbor search for inter-UAV repulsion is implemented via **decentralized neighbor discovery** over short-range V2V radio (e.g., DSRC or 5G-V2X) among UAVs sharing or crossing tubes; this avoids any centralized real-time tracking and keeps onboard latency below 10 ms.

# 6 Multi-Agent Deconfliction (Layer 4)

The trunk-and-branch architecture pre-resolves the bulk of multi-agent conflict by topology: trunk separation, flight-level assignment, and tube structure together ensure that conflicts only arise at well-defined locations (trunk junctions, station approaches, branch entries). The remaining conflicts are handled at two scales:

**Corridor-level (slow, planning-time).** Time-window allocation across overlapping nominal trajectories on the same trunk segment is solved via **Conflict-Based Search (CBS)** [10] or its bounded-suboptimal variant **ECBS** [11]. Because conflicts are sparse on a well-designed trunk graph, CBS scales effectively to the few hundred concurrent missions per trunk segment we expect.

**Drone-level (fast, online).** In-tube and at-junction conflicts at second-scale are handled by a learned decentralized policy in the spirit of **PRIMAL** [12] and GNN-based MAPF [13], augmented with the SCALED-derived local wind-aware safety envelope. This policy runs onboard and requires only local observation of nearby UAVs and the local SCALED query.

**Station approach.** Final landing sequencing at each station is a constrained queueing problem solved by **Multi-Agent Reinforcement Learning** over the small finite set of UAVs converging on a station within a 30-second window. This is a separate concern from trunk MAPF and operates on much shorter time horizons.

# 7 Trajectory Lifecycle: Incremental Updates

The nominal trajectory database is not static. The underlying cost field changes continuously: new buildings rise, seasonal wind patterns shift, vertiports come online, regulatory restrictions are amended. Aura updates the nominal database *incrementally*, asynchronously, and at a cadence appropriate to each change type:

| **Change type** | **Cadence** | **Update mechanism** |
|---|---|---|
| New building / demolition | Monthly | Incremental FMM, affected-zone only |
| Seasonal wind drift | Monthly | Full overnight FMM batch |
| Daily weather forecast | Daily | Tube width $\epsilon$ adjustment only |
| Storm / extreme weather | Hourly | Pre-computed contingency routes activated |
| New station online | On-demand | Incremental FMM for new pairs |
| Temporary NOTAM / no-fly | Minutes | Reroute via pre-computed alternates |
| In-flight wind/avoidance | Seconds | Tube MPC (local, no nominal recompute) |

The last row is qualitatively different: it does *not* update the nominal database; it executes within the existing tube via the onboard MPC. This separation of timescales is what keeps Aura tractable.

**Algorithm: Incremental Fast Marching.** When a local environmental change occurs, only the nominal trajectories that pass through (or near) the affected zone need recomputation. We use **Incremental FMM**—specifically, the techniques developed by Sethian and coauthors for fast updates of the Eikonal solution under local perturbations—together with classical incremental shortest-path algorithms (**D* Lite** [8] and **LPA*** [9]) on the discretized airspace graph. The affected-zone is identified by a domain-of-dependence analysis on the SCALED field's local update region.

**Versioning and audit.** The nominal trajectory database is version-controlled. Each UAV mission references the specific database version it was launched under. This satisfies the post-hoc auditability requirement for any safety-critical system: at any time, regulators can reconstruct which trajectory each UAV was supposed to be following, and why.

# 8 Self-Improving Fleet-Sensing Loop

Each UAV in flight is also a sensor. Onboard wind, pressure, temperature, and turbulence measurements—together with deviation statistics from the nominal trajectory—are streamed (with edge filtering) back to a central data assimilation service. These measurements update SCALED's posterior, which in turn feeds back into the nominal trajectory and tube-width computations of Section 5. This closes the loop:

$$\text{SCALED} \rightarrow \text{nominal} \rightarrow \text{fleet flies} \rightarrow \text{data} \rightarrow \text{SCALED'} \rightarrow \cdots$$

**Algorithm: data assimilation.** Updates to SCALED are performed via standard **4D-Var** [18] or **Ensemble Kalman Filter** schemes. AMCG's existing expertise in adjoint methods and data assimilation in adaptive-mesh CFD [19] provides the technical foundation; the engineering work is in adapting these to a streaming, fleet-wide ingestion pipeline.

**Avoiding sampling bias.** A naive deployment of the trunk-and-branch architecture would over-sample trunk regions and starve branches of data, leading to a self-reinforcing bias in SCALED. Aura mitigates this in three ways: (i) the asymmetric tube design, in which $\epsilon_{AB}$ grows along the direction of high SCALED uncertainty, gently steers in-tube positions toward under-sampled regions; (ii) returning UAVs (those without active forward missions) default to branch routing rather than direct trunk return; and (iii) the dual-mode operation of Section 9 actively dispatches UAVs to high-uncertainty branches during low-demand periods.

# 9 Time-Shifted Dual-Mode Fleet Operation

A central operational decision is when to use a UAV for transport versus sensing. A naive "idle-time only sensing" policy fails because demand peaks coincide with the most informative weather conditions, leaving exactly those moments uncovered. Aura adopts a two-layer policy:

**Default layer: returning-drone branch coverage.** Any UAV completing a delivery and returning to its home (or rebalancing) station defaults to a branch-routed return rather than a direct trunk return. Because the return leg is unloaded, the additional cost is small, and the cumulative effect across thousands of returns provides continuous, low-cost branch coverage at no operational sacrifice.

**Escalation layer: information-gain dispatch.** When the SCALED uncertainty in some region exceeds a threshold and the default layer's accumulated coverage is insufficient, the system actively dispatches a UAV from the transport pool to fly a sensing mission on the relevant branch. The dispatch decision is a multi-armed bandit over branches, with each branch's "arm" yielding an information gain (uncertainty reduction in SCALED) traded off against an opportunity cost (the delivery-mission revenue forgone). **Thompson sampling** or **GP-UCB** [17] suffice as off-the-shelf bandit policies; the open research question is how to specify the information-gain term jointly with the cost field of Section 5.1 so that nominal trajectories and dispatch decisions are optimized together rather than sequentially.

**Time-shifted scheduling.** At the macro scale, the fleet operates in three time-slot modes:

- **High-demand slots** (morning, evening rush): 100% transport.
- **Mid-day slots**: ∼50% transport, ∼50% sensing on rotation.
- **Late-night slots**: 100% sensing + maintenance.

This is the same operational pattern used by hub-and-spoke airlines (passenger by day, cargo by night) and is regulator-friendly because it is predictable and schedulable.

# 10 Algorithmic Specifications: Layer-by-Layer Summary

For convenience and for cross-team alignment, we summarize the algorithmic instantiation of each layer:

| Layer | Algorithm | Reference / Notes |
|---|---|---|
| Network design (L0) | Capacitated Facility Location + queueing | Standard MIP, solved via Gurobi/CPLEX |
| Mission assignment (L0.5) – strategic | 2-stage stochastic MIP | Daily/weekly cadence |
| Mission assignment (L0.5) – tactical | ALNS | Røpke & Pisinger [14], weather-aware costs |
| Mission assignment (L0.5) – reactive | Learned policy (RL) | Trained via imitation of ALNS, in spirit of [15] |
| Topological routing (L1) | Time-dependent Dijkstra/A* on trunk graph | Direct lookup against precomputed table |
| Geometric routing (L2) – offline | Anisotropic FMM | Sethian [2, 3], extending Lyu et al. [1] |
| Geometric routing (L2) – update | Incremental FMM, D* Lite, LPA* | Koenig [8, 9] |
| Kinodynamic execution (L3) | Tube MPC + funnel synthesis | Mayne [4], Majumdar [7] |
| Multi-agent (L4) – corridor | CBS / ECBS | Sharon [10], Barer [11] |
| Multi-agent (L4) – onboard | Learned decentralized (PRIMAL/GNN) | Sartoretti [12], Li [13] |
| Station approach | MARL queueing | Custom; finite small-$N$ |
| Sensing dispatch | Multi-armed bandit (TS / GP-UCB) | Srinivas [17] |
| Data assimilation (loop) | 4D-Var / EnKF | AMCG existing infra [18, 19] |

The point of this table is not novelty—most of these are textbook—but *integration discipline*: every layer in Aura has an existing, validated algorithmic anchor, so the system-level contribution is the co-design across layers rather than any single algorithmic innovation.

# 11 Additional Design Considerations

- **Heterogeneous fleet.** The system supports UAVs of different sizes, each with distinct endurance (battery life), payload capacity, and minimum turning radius. The cost field (3) is parameterized per UAV class; nominal trajectories may differ across classes for the same station pair.
- **Station throughput.** Each station has finite simultaneous launch and landing capacity, varying with station size. Inbound and outbound slot scheduling is enforced as a hard constraint at the L0/L0.5 boundary; tight slots can trigger upstream re-dispatch.
- **Battery / charging constraints.** Charging time is a hard constraint and is incorporated into the L0.5 PDP formulation as an Electric VRP variant (Schneider et al. [16]). Battery degradation across cycles is tracked and influences fleet rebalancing.
- **Failure and contingency.** For each nominal trajectory, a contingency plan (nearest emergency-landing station and the trajectory to it) is precomputed and stored alongside the nominal. Onboard the UAV, a watchdog continuously evaluates its own state against the tube; on detected anomaly, the contingency is executed without any further communication with the central system.
- **Weather-driven network reconfiguration.** Under extreme weather (storms, high winds), entire trunk segments may be temporarily closed. The system maintains a small library of pre-computed alternative trunk topologies for the most common adverse-weather patterns, allowing seconds-scale reconfiguration without replanning from scratch.

# 12 Simulation Roadmap

Validation proceeds in three progressive stages, all situated in Shenzhen, China. We additionally specify a Stage 0 to anchor the simulation infrastructure before Stage 1 expansion.

**Stage 0 — Single-corridor proof-of-concept (1 km², ∼2 weeks).** Goal: validate the offline FMM nominal trajectory generator against the SCALED-derived wind cost field for a small subregion (e.g., the area around Shenzhen Talent Park, Nanshan). Deliverables:

- One nominal trajectory map for ∼10 stations using anisotropic FMM on SCALED output;
- Visualization of trajectory vs. straight-line baseline, with energy-cost comparison;
- Verification that incremental FMM update returns identical results to full recomputation under a synthetic geometry change.

**Stage 1 — 20 km² (Nanshan District).** Goal: end-to-end transport-only simulation with all-pairs nominal trajectories, tube MPC execution, and basic CBS deconfliction. Mission types:

- Food delivery (synthetic order stream calibrated to real Meituan/ELEME density data).
- Data collection (passive on-route only, no dispatch escalation).
- Hazard/emergency response (rare-event injection to test reallocation).

**Stage 2 — 180 km² (Nanshan + Futian Districts).** Goal: introduce the trunk-and-branch hierarchy and dual-mode operation; validate the self-improving loop end-to-end.

- Food delivery with explicit order volumes (calibrated to district-level real demand).
- Information-gain dispatch for sensing missions.
- Hazard/emergency response with macro-scale rebalancing.
- Fleet-sensing data assimilated back into SCALED on a daily cadence; trajectory updates verified.

**Stage 3 — City-wide Shenzhen.** Goal: full system-level demonstration at the scale of $10^4$+ concurrent UAVs.

- Food delivery with explicit order volumes.
- Parcel delivery (multi-stop with PDP constraints).
- Data collection (active dispatch + passive coverage).
- Hazard/emergency response with city-scale rebalancing.
- Performance metrics: throughput, energy efficiency, SLA adherence, sensing coverage uniformity, safety envelope violations.

**Simulator infrastructure.** The simulator is built on a CFD-coupled discrete-event backbone: SCALED provides the wind field, the urban geometry pipeline (Faithful Contouring + OSM) provides the static environment, and a custom event-driven dispatcher manages the fleet. UAV dynamics use a standard quadrotor model (with class-conditional parameters) integrated at 100 Hz; tube MPC runs at 50 Hz; ALNS re-solves at 1 Hz; the macro-scale rebalancer runs at 0.01 Hz (every 100 s).

# 13 Open Research Questions

The Aura architecture is fully specified at the level of *which algorithm goes where*, but several research questions remain open and form natural paper exit points along the development trajectory:

1. **Tube-width design under wind uncertainty.** Formalize $\epsilon_{AB}(\mathbf{x}, t)$ as a chance-constrained optimization: minimize airspace footprint subject to $\Pr[\text{UAV exits tube}] \leq \delta$. Inputs: SCALED epistemic uncertainty, UAV dynamics, traffic mix.
2. **Learned airspace graph.** Replace the hand-designed trunk topology with a learned policy that selects trunk corridors from SCALED + geometry + demand history. Formulate as Steiner-tree-like optimization on the urban graph with learned cost.

3. **Joint nominal-and-dispatch optimization.** Currently, nominal trajectories (Section 5.1) are computed for transport efficiency, and sensing dispatch (Section 9) is computed against fixed nominals. A unified formulation in which the cost field (3) explicitly contains an information-gain term may yield a strictly Pareto-improved system.
4. **Two-layer MAPF coupling.** Trunk-level CBS and onboard learned MAPF currently operate independently. Formalize the consistency conditions between them (so that trunk-level reservations are not invalidated by onboard avoidance maneuvers).
5. **Incremental FMM under streaming SCALED updates.** What is the right cadence and granularity for nominal-trajectory updates given that SCALED itself is being continuously updated by fleet sensing? This is fundamentally a stochastic control problem on the routing infrastructure.
6. **Heterogeneous-fleet nominal trajectories.** For $K$ UAV classes and $|S|$ stations, naive storage is $O(K|S|^2)$. Are there compression schemes (e.g., low-rank decomposition over class dimension) that exploit shared structure across classes?

# 14 Conclusion

The Aura system unifies heterogeneous UAV operations under a single station-to-station abstraction, governed by a time-varying energy field inspired by $r$-adaptive mesh methods and multi-physics particle dynamics. At the macro scale, fleet rebalancing across stations follows the GMORPH analogy of grid adaptivity under a monitor function. At the micro scale, all-pairs nominal trajectories are precomputed offline via anisotropic Fast Marching on a SCALED-driven 4D wind cost field, and onboard execution is a robust tube-MPC under multi-physics potentials. The trunk-and-branch network topology, time-shifted dual-mode operation, and self-improving fleet-sensing loop together turn what would otherwise be a static routing infrastructure into a continuously self-improving platform—the same architectural pattern that underlies every successful webscale routing system. Progressive simulation from a 1 km$^2$ proof-of-concept to a city-wide deployment in Shenzhen will validate each layer in sequence.

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